

System Impact Study Report

For

***Merchant Network Upgrade Request
Queue Position AE2-009***

Nottingham – Peach Bottom Tap 230 kV

February 2020

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Spruce Valley Energy, LLC, the New Service Customer, has proposed a merchant network upgrade of the Nottingham to Peach Bottom 230 kV transmission line. The merchant network upgrade will increase the maximum emergency rating of the transmission line from existing by 11 MVA.

The proposed in-service date for this project is June 1, 2020. **This study does not imply a PECO commitment to this in-service date.**

Cost Summary

The AE2-009 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 0
Direct Connection Network Upgrades	\$ 0
Non Direct Connection Network Upgrades	\$ 0
Allocation for New System Upgrades	\$ 0
Contribution for Previously Identified Upgrades	\$ 0
Merchant Network Upgrades	\$ 1,062,000
Total Costs	\$ 1,062,000

Merchant Network Upgrades

Cost Estimates and Construction Schedule

PECO's cost estimate and schedule to complete the work is as follows:

Scope of work

- Replace Peach Bottom to Nottingham (220-08) line reactors.
- The effect of this change is to increase the summer emergency rating of the Peach Bottom to Nottingham 230 kV line by 11MVA (from 567 MVA to 578 MVA).

Cost Estimate

The cost estimate provided below is a good faith estimate for the total cost of the project. The New Service Customer will be responsible for the actual final costs for the work.

Description	Total Cost
Material	\$ 442,000
Design	\$ 45,000
Installation	\$ 120,000
Indirect Costs	\$ 238,000
Contingency	\$ 217,000
Total Costs	\$ 1,062,000

Schedule Estimate

The estimated schedule to complete the required scope of work is **30 months**.

ICTRs Granted for AE2-009:

Incremental Capacity Transfer Rights (ICTRs) were calculated using the 2022 RPM case. The following table provides the total ICTRs into the indicated Locational Deliverability Area (LDA) with the assumption that all active higher ordered queue projects were modeled in the 2022 RPM case. If there are any material changes associated with prior queued projects in the AE2 queue, then PJM would retool the analysis and determine the final ICTR's granted for the AE2-009 Merchant Transmission Request:

LDA	ICTRs
EMAAC	0
COMED	0

The New Service Customer's rights as described above shall become effective upon the completion of the Network Upgrades, Local Upgrades, or Merchant Network Upgrades, and the rights described above are contingent upon the successful completion of upgrades identified for the higher ordered queue projects. The ICTRs may change if there are changes to the higher ordered queue projects such as project withdraws, or other project modifications as described in PJM OATT §205.5. PJM would determine if a re-study is required and inform the New Service Customer.

IARRs Granted for AE2-009:

Incremental Auction Revenue Rights (IARRs) will be evaluated during the Facilities Study.